

INTERNATIONAL STANDARD

**Packaging of components for automatic handling –
Part 1: Tape packaging of components with axial leads on continuous tapes**



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INTERNATIONAL STANDARD

**Packaging of components for automatic handling –
Part 1: Tape packaging of components with axial leads on continuous tapes**

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International Standard IEC 60286-1 has been prepared by IEC technical committee 40: Capacitors and resistors for electronic equipment.

This third edition cancels and replaces the second edition published in 1997. This edition constitutes a technical revision.

This edition includes the following significant changes with respect to the previous edition:

- a) a complete revision of the structure (detailed in Annex A) and reworked layout.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
40/2538/FDIS	40/2552/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts of the IEC 60286 series, under the general title *Packaging of components for automatic handling*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

PACKAGING OF COMPONENTS FOR AUTOMATIC HANDLING –

Part 1: Tape packaging of components with axial leads on continuous tapes

1 Scope

This part of IEC 60286 applies to the tape packaging of components with axial leads for use in electronic equipment. In general, the tape is applied to the component leads.

It covers requirements for taping techniques used with equipment for the preforming of leads, automatic handling, insertion and other operations, and includes only those dimensions which are essential to the taping of components intended for the above-mentioned purposes.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60294, *Measurement of the dimensions of a cylindrical component with axial terminations*

ISO 11469, *Plastics – Generic identification and marking of plastic products*

3 Terms and definitions

No terms and definitions are listed in this document.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

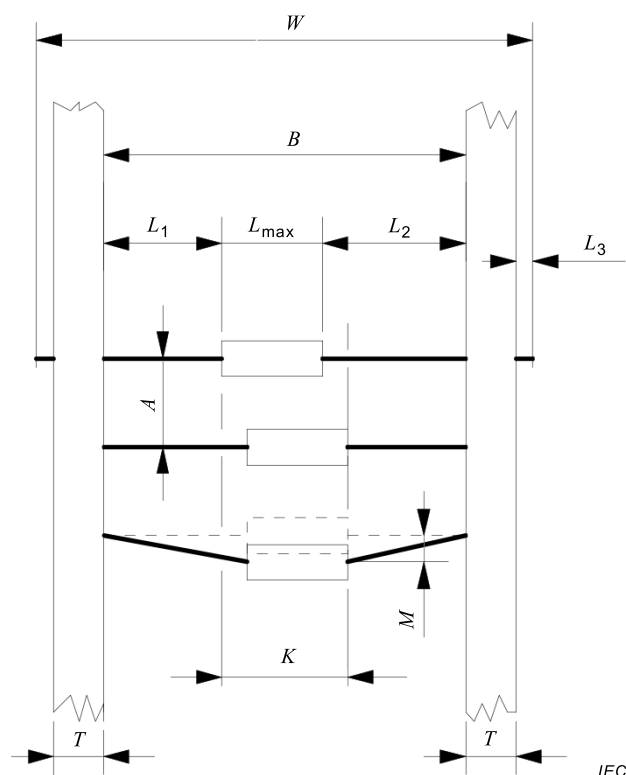
- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

4 Dimensions

4.1 Dimensions common to tapes and taped components with axial leads

4.1.1 General

The general configuration is shown in Figure 1, Table 1 and Table 2.



IEC

Key

- A is the distance between adjacent components in tape.
- B is the inner distance between two tapes.
- K is the allowed displacement distance of component in tape.
- L_1 is lead length between component body and inner side of tape on left side.
- L_2 is lead length between component body and inner side of tape on right side.
- L_3 is the part of lead protruded outside the tape.
- $L_{\max}$ is the body length of the component.
- M is the max allowed distance of displacement.
- T is the width of the tape.
- W is the maximum allowed total distance of the component after protrusion of leads outside of the tape.

Figure 1 – Dimensions for tape packaging of components with axial leads

Table 1 – Tape widths and permissible variation

Nominal widths of tape T mm	Permissible variation in width mm
6	± 1
9	± 1

Table 2 – Component spacing and permissible deviation

Standard spacing between components <i>A</i> mm	Permissible cumulative deviation over ten spacing <i>P</i> mm
5 ± 0,5	±2
10 ± 0,5	±2
15 ± 0,5	±2
20 ± 0,5	±2

4.1.2 Body location (permissible lateral deviation)

Dimension K , being the width of the window in which the component body shall be located, shall be 1,0 mm (for $B = 26$ mm) or 1,4 mm (for $B > 26$ mm) wider than the maximum length $L_{\max}$ of the component body. The latter is the body length measured in accordance with IEC 60294.

Unless otherwise specified, the window shall be centrally located between the tapes.

When a gauge measurement to determine dimension K is impractical, it is possible to measure the distances L_1 and L_2 of the leads. The difference between L_1 and L_2 shall not exceed 1,0 mm (for $B = 26$ mm) or 1,4 mm (for $B > 26$ mm).

4.1.3 Tape spacing

The maximum overall width W , including any lead protrusion L_3 , shall not exceed 140 mm. Its value shall be chosen in relation to the component dimensions.

The tape spacing given in Table 3 is recommended.

Table 3 – Tape spacing

Inside tape spacing <i>B</i> mm
26 $^{+1,5}_0$
52 $^{+2}_{-1}$
63 ± 2
73 ± 2
83 ± 2
93 ± 2
Inside tape spacing B above 93 mm to be in 10 mm increments with a tolerance ±2 mm as long as the maximum overall width W of 140 mm is not exceeded.

4.2 Taping**4.2.1 General**

The following requirements for axial components shall be met as appropriate (see Figure 1).

4.2.2 Orientation of polarized components

All polarized components shall be oriented in one direction; the polarized components shall be clearly identified by use of different colours or marked tapes, the anode being of a white or neutral colour.

4.2.3 Position, kinks and bends of the leads

The position, kinks and bends of the leads are as follows.

- a) The leads shall be free from kinks and bends.
- b) The method used to position the component leads on the tape shall be such that the leads are not nicked or otherwise damaged.
- c) Leads shall not be bent beyond 1,0 mm (for $B = 26$ mm) or 1,2 mm (for $B > 26$ mm) from their nominal position when measured from the leading edge of the component lead at the inside tape edge and at the lead egress from the component (see dimension M in Figure 1).

4.2.4 The ends of the leads

The ends of the leads shall not protrude beyond the tapes.

4.2.5 Holding in the tape

The components shall be held sufficiently in the tape(s) so that they cannot come free during normal handling.

4.2.6 Requirements of tape material, storage, and handling

The tapes shall be suitable to withstand storage of the taped components. The tape material shall not migrate along the leads or give off vapours that can affect solderability or deteriorate the component properties or leads by chemical action (e.g. corrosion).

In addition, the tapes shall not become detached, thereby causing the components to lose their relative position after storage, and the tape shall not age to such an extent that its strength is reduced, causing it to break either when the components are unreeled manually or by an assembly machine.

Tapes in adjacent layers shall not stick together in the packing, because of the exposed adhesive for instance.

4.2.7 Splices of tape

Splices shall be at least as strong as the original tape and shall not be thicker than four times the thickness of a single layer of the original tape. Splices shall not be misaligned by more than 0,8 mm. When splicing is applied, component spacing A shall remain within the tolerances as specified in Table 2.

4.2.8 Leader tape

A twin leader tape, free of components, having a minimum length of 200 mm shall be provided at the beginning and at the end of the tape.

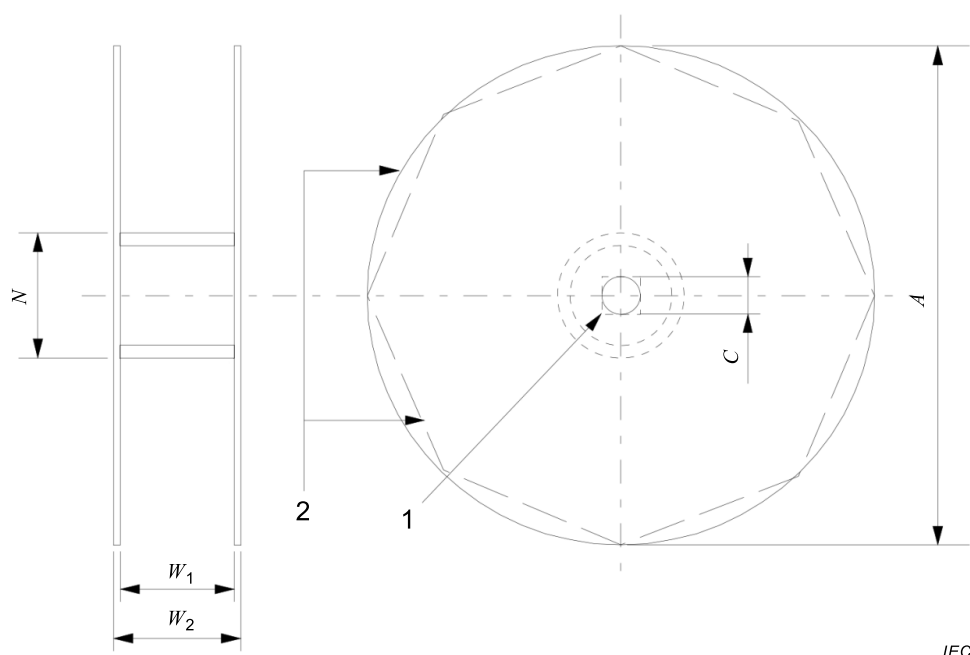
5 Packing

5.1 General

The tapes of components may either be wound on reels or fan-folded.

5.2 Dimensions of the reel

The preferred reel dimensions are shown in Figure 2.



Key

- 1 is the optional shape: circular or square.
- 2 is the optional shape: circular or polygonal.
- A is the reel diameter: maximum 400 mm.
- C is the arbor hole diameter: 14 mm to 38 mm.
- N is the hub diameter: 34,9 mm to 102 mm.

Figure 2 – Reel dimensions

5.3 Distances W_1 and W_2

The distances W_1 and W_2 shall be governed by the overall width of the taped components W (see Figure 1) and shall allow proper reeling and unreeling.

5.4 Recycling

Reels as defined in Figure 2 should be made of recyclable material. When such material is used, the reel shall be permanently marked with the recycling symbol.

ISO 11469 should be used for making the material.

5.5 Protection of components

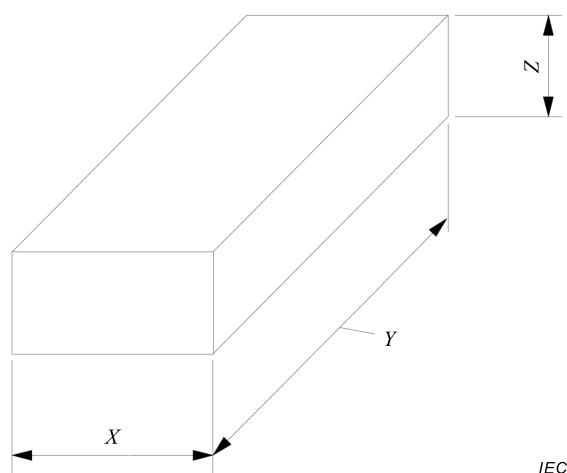
In order to prevent component damage and lead distortion, protection between layers of components and over the last layer may be necessary. In this case, protection materials shall not cause deterioration of the component or of lead solderability.

5.6 Reel filling

The total number of components on the reel shall be such that the components and the final cover shall not extend beyond the smallest dimension of the flange in the radial direction.

5.7 Dimensions of the fan-fold container

The preferred outer dimensions of a fan-fold arrangement are shown in Figure 3.



Key

- X* is the width of the fan-folded arrangement: controlled by *W*.
- Y* is the length of the fan-folded arrangement: maximum 610 mm.
- Z* is the height of the fan-folded arrangement: maximum 150 mm.

Figure 3 – Outer dimensions of the fan-fold arrangement

5.8 Missing components

For automatic insertion, it is required that the number of empty places in the tape, per reel or fan-fold arrangement, shall not exceed 0,1 % or one position, whichever is greater, without consecutive empty places, unless agreed otherwise between the customer and the manufacturer.

6 Marking

When required, the tape shall be marked at every 50th or 100th component place.

Annex A

(informative)

Cross-reference for references to the previous edition of this document

The revision of this document has resulted in new numbering for some clauses. Table A.1 provides cross-references for the clause numbering of this document compared to its previous edition.

Table A.1 – Cross-reference to clauses

IEC 60286-1:1997 2nd edition	IEC 60286-1:2017 3rd edition	Notes
Clause	Clause	
1	1	General and scope are merged into one
1.1		
1.2	2	In accordance with the ISO/IEC Directives – Part 2
—	3	
2	4	In accordance with the change of clause numbers
3	5	
4	6	

Bibliography

IEC 60286-2, *Packaging of components for automatic handling – Part 2: Tape packaging of components with unidirectional leads on continuous tapes*

IEC 60286-3, *Packaging of components for automatic handling – Part 3: Packaging of surface mount components on continuous tapes*

IEC 60301, *Preferred diameters of wire terminations of capacitors and resistors*

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